Lab 7: Fuzzing for Input Validation Bugs

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Section: Sem-5, Section J

Step 2: Output of the buggy processor.py

```
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$ python
processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 130,85,39.5
Summary: Vitals: HR=130.0, SpO2=85.0% - CRITICAL

==== Running compute_patient_bmi ====
Enter weight/height (e.g. 70/1.75): 70/1.75
BMI: 22.86

==== Running assign_ward_bed ====
Enter allocation (e.g. ICU-05): ICU-05
Assignment: Intensive Care Unit Bed #5
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$
```

```
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$ python
processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): abc
Traceback (most recent call last):
  File "/home/sarvani/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 41, in <modu
le>
    main()
    ~~~~^^
  File "/home/sarvani/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 30, in main
    print("Summary:", parse_vitals_summary(v_input))
                      ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^
  File "/home/sarvani/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 3, in parse_
vitals_summary
    hr = float(parts[0])
ValueError: could not convert string to float: 'abc'
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$
```

Step 4: test_processor.py failing on the buggy code

```
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/sarvani/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 3
2, in test_assign_ward_bed_safe
    |     assert False, f"assign_ward_bed crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: assign_ward_bed crashed on '' with: ValueError: not enough values to unp
ack (expected 2, got 1)
    | assert False
    | Failing test case: test_assign_ward_bed_safe(
    |     s='',  # or any other generated value
    | )
    +--------------- 3 ---------------
    | Traceback (most recent call last):
    |   File "/home/sarvani/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 2
9, in test_assign_ward_bed_safe
    |     result = assign_ward_bed(s)
    |   File "/home/sarvani/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 17, in
 assign_ward_bed
    |     ward, bed_str = allocation_code.strip().split('-')
    |                     ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/sarvani/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 3
2, in test_assign_ward_bed_safe
    |     assert False, f"assign_ward_bed crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: assign_ward_bed crashed on None with: AttributeError: 'NoneType' object
has no attribute 'strip'
    | assert False
    | Failing test case: test_assign_ward_bed_safe(
    |     s=None,  # or any other generated value
    | )
    +---------------------------------
=================================== short test summary info ====================================
FAILED test_processor.py::test_parse_vitals_summary_no_crash - ExceptionGroup: Hypothesis foun
d 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_compute_patient_bmi_safe - ExceptionGroup: Hypothesis found 3 d
istinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_assign_ward_bed_safe - ExceptionGroup: Hypothesis found 3 disti
nct failures. (3 sub-exceptions)
==================================== 3 failed in 6.01s =====================================
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$
```

Step 5: Output of the fixed processor.py

```
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$ python
processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 130,85,39.5
Summary: Vitals: HR=130.0, SpO2=85.0% - CRITICAL

==== Running compute_patient_bmi ====
Enter weight/height (e.g. 70/1.75): 70/1.75
BMI: 22.86

==== Running assign_ward_bed ====
Enter allocation (e.g. ICU-05): ICU-05
Assignment: Intensive Care Unit Bed #5
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$
```

```
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$ python
processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): abc
Summary: Invalid input: expected exactly 3 values (HR,SpO2,Temp)

==== Running compute_patient_bmi ====
Enter weight/height (e.g. 70/1.75): 70/0
BMI: Invalid input: weight and height must be greater than 0

==== Running assign_ward_bed ====
Enter allocation (e.g. ICU-05): XYZ-9
Assignment: Invalid input: unknown ward (use ER, ICU, GEN or PED)
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$
```

Step 6: test_processor.py passing on the fixed code

```
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$ pytest
test_processor.py
================================== test session starts ==================================
platform linux -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: /home/sarvani/L6_L7_07_hospital_traige_priority/Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_processor.py ...                                                            [100%]

=================================== 3 passed in 0.64s ===================================
(.venv) sarvani@sarvani-Aspire-Lite-AL15-41:~/L6_L7_07_hospital_traige_priority/Lab 7$
```